Engineering Economy

Chapter 2:
Cost Concepts and Design Economics

Chapter 2
Cost concept and design economics

The objective is
to analyze short-term alternatives when the time value of money is not a factor.

Chapter 2
2.1. Cost Terminology

There are variety of costs in an engineering economic analysis

Costs differ in their

- Frequency of occurance
- Relative magnitude
- Degree of impact on the study

Chapter 2
2.1. Cost Terminology

Costs can be categorized in several different ways

- *Fixed cost*
- *Variable cost*
- *Incremental cost*

- *Cash cost*
- *Book cost*
- *Sunk cost*

- *Direct cost*
- *Indirect cost*
- *Standard cost*

- *Opportunity cost*
- *Life cycle cost*

Chapter 2
2.1. Cost Terminology

- ***Fixed costs***

unaffected by changes in activity level

tend to remain constant

affected when larger changes in usage of resources occur

affected when plant expansion or shutdown is involved

Typical fixed costs are:

insurance and taxes on facilities,

general management and administrative salaries,

license fees,

interest costs on borrowed capital

Chapter 2
2.1. Cost Terminology

- ***Variable cost*s**

associated with an operation

vary in total with the quantity of output (or similar measure of activity)

Typical variable costs are:

material used in a product or service

labor used in a product or service

Chapter 2
2.1. Cost Terminology



Chapter 2
2.1. Cost Terminology

- ***Incremental costs***

additional cost resulting from increasing output of a system by one (or more) units

involve a limited change in output or activity level

calculated by analyzing the additional expenses involved in production process

Chapter 2
Example 2.1

Fixed and Variable Costs

In connection with surfacing a new highway, a contractor has a choice of two sites on which to set up the asphalt-mixing plant equipment. The contractor estimates that it will cost \$2.75 per cubic yard mile (yd^3-mile) to haul the asphalt-paving material from the mixing plant to the job location. Factors relating to the two mixing sites are as follows (production costs at each site are the same):

Chapter 2
Example 2.1

Fixed and Variable Costs

Cost Factor	Site *A*	Site *B*
Average hauling distance	4 miles	3 miles
Monthly rental of site	$2,000	$7,000
Cost to set up and remove equipment	$15,000	$50,000
Hauling expense	$2.75/yd^3-mile	$2.75/yd^3-mile
Flagperson	Not required	$150/day

The job requires 50,000 cubic yards of mixed-asphalt-paving material. It is estimated that four months (17 weeks of five working days per week) will be required for the job. Compare the two sites in terms of their fixed, variable, and total costs. Assume that the cost of the return trip is negligible. Which is the better site? For the selected site, how many cubic yards of paving material does the contractor have to deliver before starting to make a profit if paid $12 per cubic yard delivered to the job location?

Chapter 2
Example 2.1

EXAMPLE 2-1 **Fixed and Variable Costs**

Solution

The fixed and variable costs for this job are indicated in the table shown next. Site rental, setup, and removal costs (and the cost of the flagperson at Site *B*) would be constant for the total job, but the hauling cost would vary in total amount with the distance and thus with the total output quantity of yd^3-miles (x).

Cost	Fixed	Variable	Site *A*	Site *B*
Rent	√		= $8,000	= $28,000
Setup/removal	√		= 15,000	= 50,000
Flagperson	√		= 0	5(17)($150) = 12,750
Hauling		√	4(50,000)($2.75) = 550,000	3(50,000)($2.75) = 412,500
			Total: $573,000	$503,250

Chapter 2
Example 2.1

EXAMPLE 2-1 **Fixed and Variable Costs**

Site *B*, which has the larger fixed costs, has the smaller total cost for the job. Note that the extra fixed costs of Site *B* are being "traded off" for reduced variable costs at this site.

The contractor will begin to make a profit at the point where total revenue equals total cost as a function of the cubic yards of asphalt pavement mix delivered. Based on Site *B*, we have

$$3(\$2.75) = \$8.25 \text{ in variable cost per yd}^3 \text{ delivered}$$

$$\text{Total cost} = \text{total revenue}$$

$$\$90{,}750 + \$8.25x = \$12x$$

$$x = 24{,}200 \text{ yd}^3 \text{ delivered.}$$

Therefore, by using Site *B*, the contractor will begin to make a profit on the job after delivering 24,200 cubic yards of material.

Chapter 2
2.1. Cost Terminology

More ways to categorize costs

- ***Direct costs***

can be measured and allocated to a specific work activity

costs directly associated with a good/service are direct costs

For example, materials needed to make a pair of scissors would be a direct cost.

Chapter 2
2.1. Cost Terminology

More ways to categorize costs

- ***Indirect costs***

difficult to attribute or allocate to a specific output or work activity (also *overhead* or *burden*)

allocated through, a selected formula to the outputs

For example, the costs of common tools, genereal supplies, and equipment maintenance in a plant are treated as indirect costs.

Chapter 2
2.1. Cost Terminology

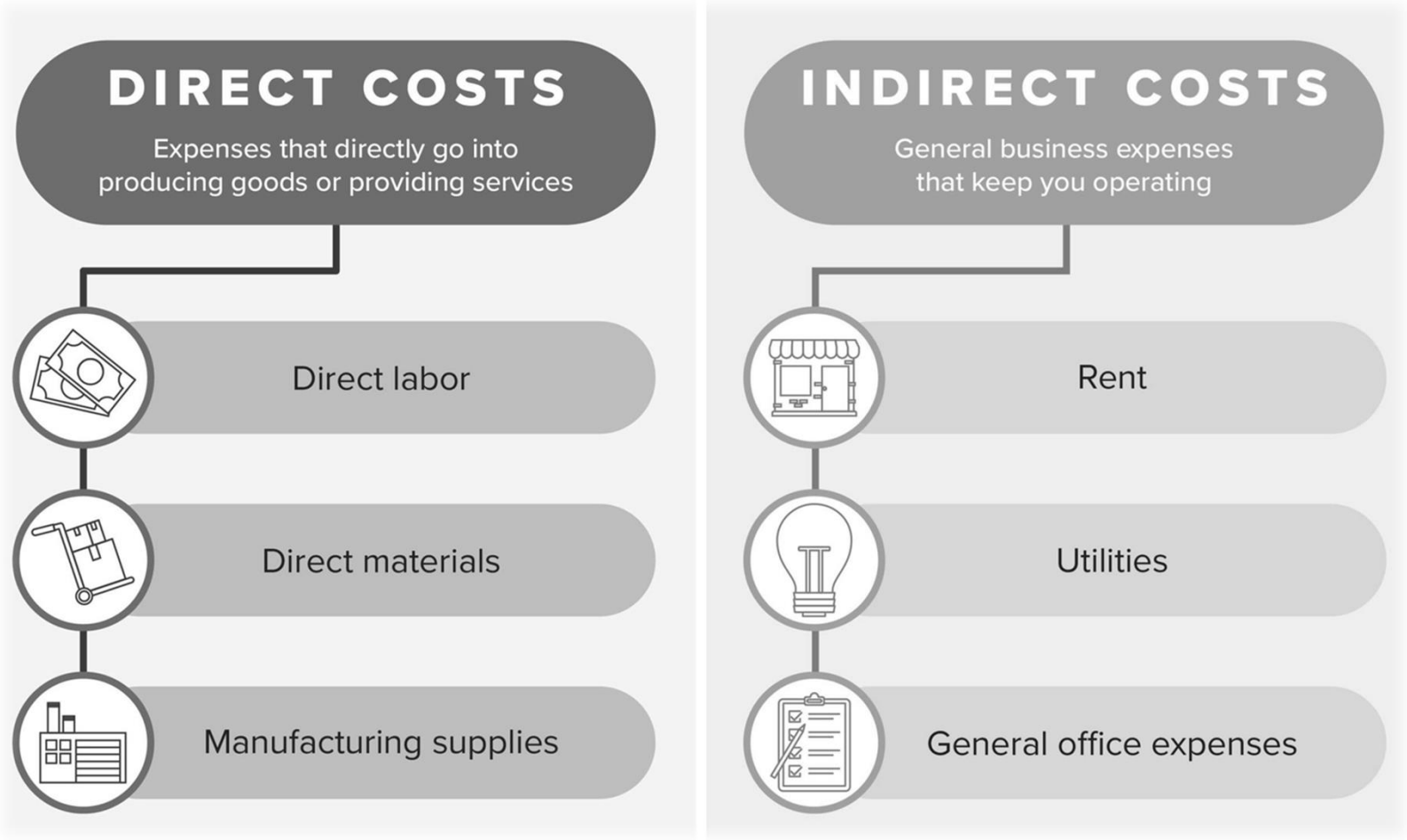
More ways to categorize costs

- ***Standard costs***

planned cost per unit of output established in advance of production or service delivery

developed from anticipated direct labor hours, materials, and overhead categories

Chapter 2
2.1. Cost Terminology



Chapter 2
2.1. Cost Terminology

We need to use common cost terminology

- ***Cash costs***

a cost that involves a payment of cash and results in a cash flow

- ***Book costs***

a cost that does not involve a cash transaction but is reflected in the accounting system

- ***Sunk costs***

a cost that has occurred in the past and has no relevance to estimates of future costs and revenues related to an alternative course of action.

Sunk costs are irretrievable consequences of past decisions and therefore irrelevant in the analysis and comparison of alternatives

Chapter 2
Example 2.2

Sunk Costs in Replacement Analysis

A classic example of sunk cost involves the replacement of assets. Suppose that your firm is considering the replacement of a piece of equipment. It originally cost $50,000, is presently shown on the company records with a value of $20,000, and can be sold for an estimated $5,000. For purposes of replacement analysis, the $50,000 is a sunk cost. However, one view is that the sunk cost should be considered as the difference between the value shown in the company records and the present realizable selling price. According to this viewpoint, the sunk cost is $20,000 minus $5,000, or $15,000. Neither the $50,000 nor the $15,000, however, should be considered in an engineering economic analysis, except for the manner in which the $15,000 may affect income taxes, which will be discussed in Chapter 9.

Chapter 2
2.1. Cost Terminology

More common cost terminology

- ***Opportunity cost***

the monetary advantage foregone due to limited resources, the cost of the best rejected opportunity,

is often hidden or implied.

For example;

Consider a student who could earn $20,000 for working during a year, but choses instead to go to school for a year and spend $5,000 to do so.

The opportunity cost of going to school for that year is $25,000 ($5,000 cash outlay + $20,000 for income forgone)

Chapter 2
Example 2.3

Opportunity Cost in Replacement Analysis

The concept of an opportunity cost is often encountered in analyzing the replacement of a piece of equipment or other capital asset. Let us reconsider Example 2-2, in which your firm considered the replacement of an existing piece of equipment that originally cost $50,000, is presently shown on the company records with a value of $20,000, but has a present market value of only $5,000. For purposes of an engineering economic analysis of whether to replace the equipment, the present investment in that equipment should be considered as $5,000, because, by keeping the equipment, the firm is giving up the *opportunity* to obtain $5,000 from its disposal. Thus, the $5,000 immediate selling price is really the investment cost of not replacing the equipment and is based on the opportunity cost concept.

Chapter 2
2.1. Cost Terminology

More common cost terminology

- ***Life-cycle costs***

the summation of all costs related to a product, structure, system, or service during its life span.

the life cycle begins with identification of the economic need or want and ends with retirement and disposal activities.

Basic life cycle cost categories are:

investment cost

operation and maintenance cost

disposal cost

Chapter 2
2.2. The General Economic Concepts

2.2.1. Consumer and producer goods and services

Consumer goods and services are those products or services that are directly used by people to satisfy their wants.

Food, clothing, homes, cars, medical services, tv, etc.

Producer goods and services are used to produce consumer goods and services or other producer goods.

Machine tools, factory buildings, buses, etc

Chapter 2
2.2. The General Economic Concepts

2.2.2. Necessities, Luxuries, and Price Demand

Goods and services may be divided into two types: necessities and luxuries.

These terms are relative, because, for most goods and services, what one person considers a necessity may be considered a luxury by another.

For all goods and services, there is a relationship between the price that must be paid and the quantity that will be demanded or purchased.

Chapter 2
2.2. The General Economic Concepts

Price - Demand

$$p = a - b\mathrm{D} \qquad \text{(Eq.2.1)}$$

$$\text{for} \quad 0 \leq D \leq \frac{a}{b} \quad \text{and} \quad a > 0, \quad b > 0$$

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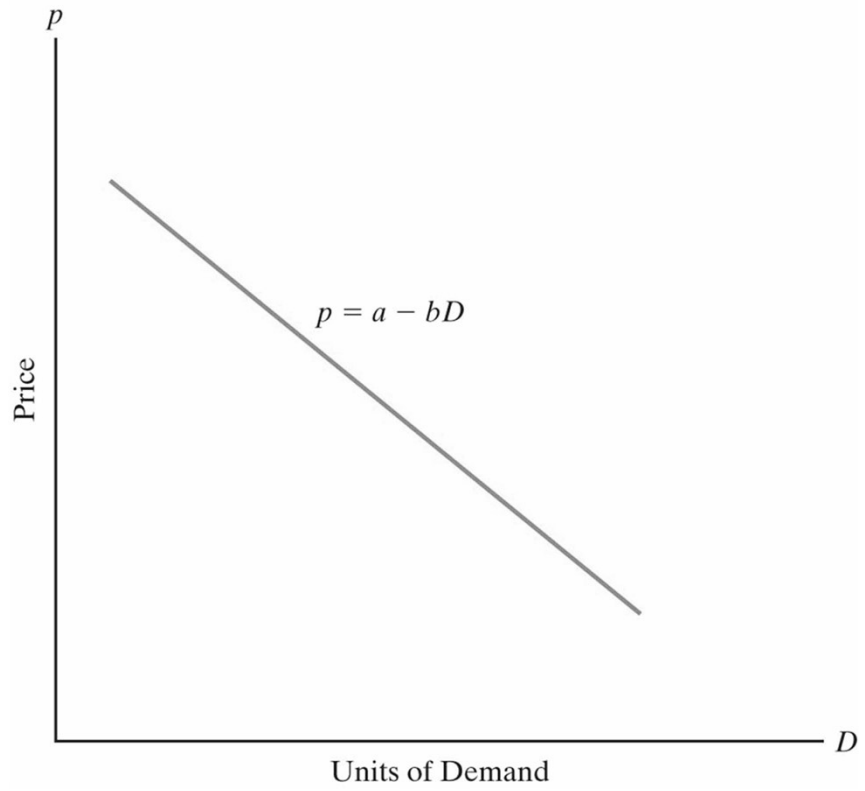
Chapter 2
2.2. The General Economic Concepts

The general price-demand relationship

The demand for a product or service is directly related to its price according to

$$p = a - bD$$

where p is price, D is demand, and a and b are constants that depend on the particular product or service.



Chapter 2
2.2. The General Economic Concepts

2.2.3. The Total Revenue Function

The total revenue is the amount received from the sale of the product. *So we can formulize this definition as:*

$$TR = \text{price} \times \text{demand} = p \times D \qquad \text{(Eq.2.2)}$$

If we use the relationship between price and demand as given in Eq.2.1.

$$TR = p \times D$$
$$p = a - bD$$
$$TR = (a - bD) \times D$$
$$TR = aD - bD^2$$

$a > 0$ and $b > 0$, $0 < D < a/b$

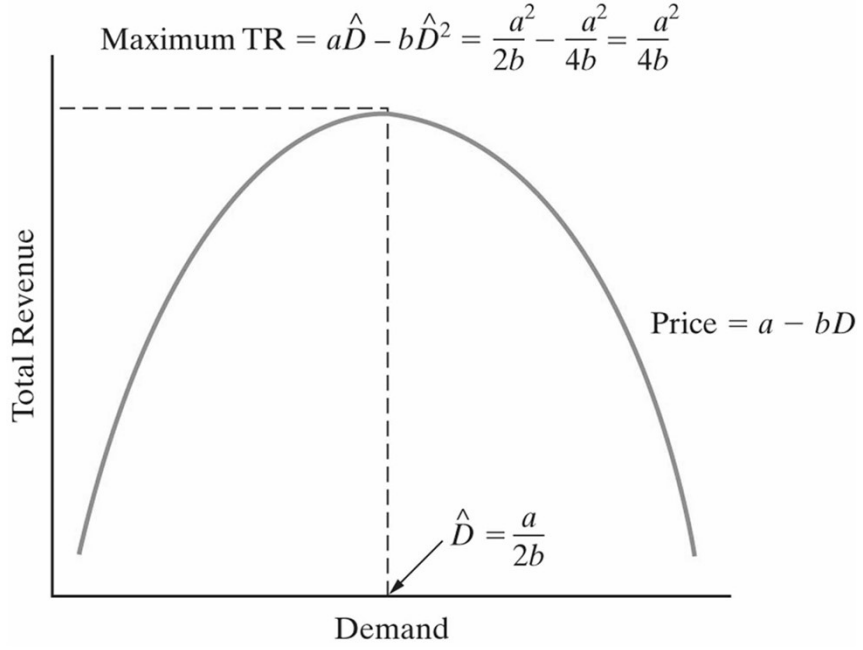
Chapter 2
2.2. The General Economic Concepts

Calculus can help determine the demand that maximizes revenue.

$$\frac{dTR}{dD} = a - 2bD = 0$$

Solving, the optimal demand is (Eq. 2.3)

$$\hat{D} = \frac{a}{2b}$$



Chapter 2
2.2. The General Economic Concepts

$$TR_{max} = a\hat{D} - b\hat{D}^2$$

$$TR_{max} = a\frac{a}{2b} - b\left(\frac{a}{2b}\right)^2$$

$$TR_{max} = \frac{a^2}{2b} - \frac{a^2}{4b}$$

$$TR_{max} = \frac{a^2}{4b} \qquad \text{(Eq.2.4.)}$$

Chapter 2
2.2. The General Economic Concepts

2.2.3. Cost, Volume, and Breakeven Point Relationships

As we previously discussed;

$C_T = C_F + C_V$

C_F and C_V denote fixed and variable costs, respectively.

For the linear relationship assumed here:

$C_V = c_v \times D$

c_v denotes the variable cost per unit

$$\text{Profit} = TR - TC$$
$$= (aD - bD^2) - (C_F - c_vD)$$

$$\text{Profit} = -bD^2 + (a-c_v)D - C_F \qquad \text{(Eq.2.5)}$$

Chapter 2
2.2. The General Economic Concepts

In order for a profit to occur two conditions must be met :

1- $(a-c_v)>0$; that is, the price per unit that will result in no demand has to be greater than the variable cost per unit (this avoids negative demand)

2- Total revenue must exceed total cost

If these conditions are met, we can find the optimal demand at which max profit will occur by taking the first derivative of E.q.2.5 with respect to D and setting it equal to zero

Chapter 2
2.2. The General Economic Concepts

2.2.3. Cost, Volume, and Breakeven Point Relationships

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$$\text{Profit} = -bD^2 + (a-c_v)D - C_F \qquad \text{(Eq.2.5)}$$

Chapter 2
2.2. The General Economic Concepts

We can also find maximum profit…

Profit is revenue minus cost, so

$$\text{Profit} = -bD^2 + (a - c_v)D - C_F$$

for

$$0 \leq D \leq \frac{a}{b} \quad \text{and} \quad a > 0, b > 0$$

Differentiating, we can find the value of D that maximizes profit (eq. 2-6).

$$D^* = \frac{a - c_v}{2b}$$

Chapter 2
2.2. The General Economic Concepts

An economic breakeven point for an operation occurs when total revenue equals to total cost

Total revenue = Total cost

$aD - bD^2 = C_F - c_vD$

$-bD^2 + (a-c_v)D - C_F = 0$

Because this equation is a quadratic equation with one unknown (D), we can solve for the breakeven points D1' and D2' (the roots of the equation)

Chapter 2
2.2. The General Economic Concepts

Remember that a quadratic equation:

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \mp \sqrt{(b^2 - 4ac}}{2a}$$

Therefore;

$$-bD^2 + (a-c_v)D - C_F = 0$$

is a quadratic equation with one unknown

So, solution is;

x » D

b » $(a-c_v)$

a » -b

c » $-C_F$

$$D' = \frac{-(a-c_v) \pm \sqrt{(a-c_v)^2 - 4(-b)(-C_F)}}{2(-b)}$$

Chapter 2
2.2. The General Economic Concepts

And we can find revenue/cost breakeven.

Breakeven is found when total revenue = total cost.

$$D' = \frac{-(a - c_v) \pm \sqrt{(a - c_v)^2 - 4(-b)(-C_F)}}{2(-b)}$$

Chapter 2
Example

Acme Manufacturing is a major player in the lawn sprinkler business. Their high-end sprinkler is used commercially, and is quite popular with golf course greens keepers.

In producing these sprinklers Acme's fixed cost (C_F) is \$55,000 per month with a variable cost (c_v) of \$15.50 per unit.

The selling price for these high-end sprinklers is described by the equation p=\$87.50 – 0.02(D).

a)What is the optimal volume of sprinklers? Does Acme make a profit at that volume?

b)What is the range of profitable demand?

Chapter 2
Example

a) What is the optimal volume of sprinklers? Does Acme make a profit at that volume?

$$D^* = \frac{a - c_v}{2b} = \frac{87.50 - 15.50}{2(0.02)} = \$1,800$$

$$\text{Profit} = -(0.02)(1,800)^2 + (72)(1,800) - 55,000 = \$9,800$$

b) What is the range of profitable demand?

$$D' = \frac{-(a - c_v) \pm \sqrt{(a - c_v)^2 - 4(-b)(-C_F)}}{2(-b)}$$

$$D'_1 = 1,100 \qquad\qquad D'_2 = 2,500$$

Chapter 2
Example 2-4

Optimal Demand When Demand Is a Function of Price

A company produces an electronic timing switch that is used in consumer and commercial products. The fixed cost (C_F) is \$73,000 per month, and the variable cost (c_v) is \$83 per unit. The selling price per unit is $p = \$180 - 0.02(D)$, based on Equation (2-1). For this situation,

(a) determine the optimal volume for this product and confirm that a profit occurs (instead of a loss) at this demand.

(b) find the volumes at which breakeven occurs; that is, what is the range of profitable demand? Solve by hand and by spreadsheet.

Chapter 2
Solution 2-4

(a) $D^* = \dfrac{a - c_v}{2b} = \dfrac{\$180 - \$83}{2(0.02)} = 2{,}425$ units per month [from Equation (2-10)].

Is $(a - c_v) > 0$?

$$(\$180 - \$83) = \$97, \quad \text{which is greater than } 0.$$

And is (total revenue − total cost) > 0 for $D^* = 2{,}425$ units per month?

$$[\$180(2{,}425) - 0.02(2{,}425)^2] - [\$73{,}000 + \$83(2{,}425)] = \$44{,}612$$

A demand of $D^* = 2{,}425$ units per month results in a maximum profit of \$44,612 per month. Notice that the second derivative is negative (−0.04).

Chapter 2
Solution 2-4

(b) Total revenue = total cost (breakeven point)

$$-bD^2 + (a - c_v)D - C_F = 0 \qquad \text{[from Equation (2-11)]}$$

$$-0.02D^2 + (\$180 - \$83)D - \$73{,}000 = 0$$

$$-0.02D^2 + 97D - 73{,}000 = 0$$

And, from Equation (2-12),

$$D' = \frac{-97 \pm [(97)^2 - 4(-0.02)(-73{,}000)]^{0.5}}{2(-0.02)}$$

$$D'_1 = \frac{-97 + 59.74}{-0.04} = 932 \text{ units per month}$$

$$D'_2 = \frac{-97 - 59.74}{-0.04} = 3{,}918 \text{ units per month.}$$

Thus, the range of profitable demand is 932–3,918 units per month.

Chapter 2
Example 2-5

Breakeven Point When Price Is Independent of Demand

An engineering consulting firm measures its output in a standard service hour unit, which is a function of the personnel grade levels in the professional staff. The variable cost (c_v) is \$62 per standard service hour. The charge-out rate [i.e., selling price (p)] is \$85.56 per hour. The maximum output of the firm is 160,000 hours per year, and its fixed cost (C_F) is \$2,024,000 per year. For this firm,

(a) what is the breakeven point in standard service hours and in percentage of total capacity?

(b) what is the percentage reduction in the breakeven point (sensitivity) if fixed costs are reduced 10%; if variable cost per hour is reduced 10%; and if the selling price per unit is increased by 10%?

Chapter 2
Solution 2-5

Breakeven Point When Price Is Independent of Demand

Solution

(a)

$$\text{Total revenue} = \text{total cost} \qquad \text{(breakeven point)}$$

$$pD' = C_F + c_v D'$$

$$D' = \frac{C_F}{(p - c_v)}, \tag{2-13}$$

and

$$D' = \frac{\$2{,}024{,}000}{(\$85.56 - \$62)} = 85{,}908 \text{ hours per year}$$

$$D' = \frac{85{,}908}{160{,}000} = 0.537,$$

or 53.7% of capacity.

Chapter 2
Solution 2-5

Breakeven Point When Price Is Independent of Demand

(b) A 10% reduction in C_F gives

$$D' = \frac{0.9(\$2{,}024{,}000)}{(\$85.56 - \$62)} = 77{,}318 \text{ hours per year}$$

and

$$\frac{85{,}908 - 77{,}318}{85{,}908} = 0.10,$$

or a 10% reduction in D'.

A 10% reduction in c_v gives

$$D' = \frac{\$2{,}024{,}000}{[\$85.56 - 0.9(\$62)]} = 68{,}011 \text{ hours per year}$$

and

$$\frac{85{,}908 - 68{,}011}{85{,}908} = 0.208,$$

Chapter 2
Solution 2-5

Breakeven Point When Price Is Independent of Demand

or a 20.8% reduction in D'.

A 10% increase in p gives

$$D' = \frac{\$2{,}024{,}000}{[1.1(\$85.56) - \$62]} = 63{,}021 \text{ hours per year}$$

and

$$\frac{85{,}908 - 63{,}021}{85{,}908} = 0.266,$$

or a 26.6% reduction in D'.

Thus, the breakeven point is more sensitive to a reduction in variable cost per hour than to the same percentage reduction in the fixed cost. Furthermore, notice that the breakeven point in this example is highly sensitive to the selling price per unit, p.

Chapter 2
2.3. Cost-Driven Design Optimization

Engineers must consider cost in the design of products, processes and services.

- “Cost-driven design optimization” is critical in today’s competitive business environment.
- In our brief examination we examine discrete and continuous problems that consider a single primary cost driver.

Chapter 2
2.3. Cost-Driven Design Optimization

Two main tasks are involved in cost-driven design optimization.

1. Determine the optimal value for a certain alternative's design variable.

2. Select the best alternative, each with its own unique value for the design variable.

Cost models are developed around the design variable, X.

Chapter 2
2.3. Cost-Driven Design Optimization

Optimizing a design with respect to cost is a four-step process.

- Identify the design variable that is the primary cost driver.
- Express the cost model in terms of the design variable.
- For continuous cost functions, differentiate to find the optimal value. For discrete functions, calculate cost over a range of values of the design variable.
- Solve the equation in step 3 for a continuous function. For discrete, the optimum value has the minimum cost value found in step 3.

Chapter 2
2.3. Cost-Driven Design Optimization

Here is a simplified cost function.

$$\text{Cost} = aX + \frac{b}{X} + k$$

where,

a is a parameter that represents the directly varying cost(s),

b is a parameter that represents the indirectly varying cost(s),

k is a parameter that represents the fixed cost(s), and

X represents the design variable in question.

Chapter 2
2.3. Cost-Driven Design Optimization

"Present economy studies" can ignore the time value of money.

- Alternatives are being compared over one year or less.
- When revenues and other economic benefits vary among alternatives, choose the alternative that maximizes overall profitability of defect-free output.
- When revenues and other economic benefits are not present or are constant among alternatives, choose the alternative that minimizes total cost per defect-free unit.

Chapter 2
Example 2-6

How Fast Should the Airplane Fly?

The cost of operating a jet-powered commercial (passenger-carrying) airplane varies as the three-halves (3/2) power of its velocity; specifically, $C_O = knv^{3/2}$, where n is the trip length in miles, k is a constant of proportionality, and v is velocity in miles per hour. It is known that at 400 miles per hour, the *average* cost of operation is \$300 per mile. The company that owns the aircraft wants to minimize the cost of operation, but that cost must be balanced against the cost of the passengers' time (C_C), which has been set at \$300,000 per hour.

(a) At what velocity should the trip be planned to minimize the total cost, which is the sum of the cost of operating the airplane and the cost of passengers' time?

(b) How do you know that your answer for the problem in Part (a) minimizes the total cost?

Chapter 2
Solution 2-6

How Fast Should the Airplane Fly?

Solution

(a) The equation for total cost (C_T) is

$$C_T = C_O + C_C = knv^{3/2} + (\$300{,}000 \text{ per hour})\left(\frac{n}{v}\right),$$

where n/v has time (hours) as its unit.

Now we solve for the value of k:

$$\frac{C_O}{n} = kv^{3/2}$$

$$\frac{\$300}{\text{mile}} = k\left(400\frac{\text{miles}}{\text{hour}}\right)^{3/2}$$

$$k = \frac{\$300/\text{mile}}{\left(400\frac{\text{miles}}{\text{hour}}\right)^{3/2}}$$

$$k = \frac{\$300/\text{mile}}{8000\left(\frac{\text{miles}^{3/2}}{\text{hour}^{3/2}}\right)}$$

$$k = \$0.0375\frac{\text{hours}^{3/2}}{\text{miles}^{5/2}}.$$

Chapter 2
Solution 2-6

How Fast Should the Airplane Fly?

Thus,

$$C_T = \left(\$0.0375 \frac{\text{hours}^{3/2}}{\text{miles}^{5/2}}\right)(n \text{ miles})\left(v \frac{\text{miles}}{\text{hour}}\right)^{3/2} + \left(\frac{\$300{,}000}{\text{hour}}\right)\left(\frac{n \text{ miles}}{v \frac{\text{miles}}{\text{hour}}}\right)$$

$$C_T = \$0.0375 n v^{3/2} + \$300{,}000\left(\frac{n}{v}\right).$$

Next, the first derivative is taken:

$$\frac{dC_T}{dv} = \frac{3}{2}(\$0.0375) n v^{1/2} - \frac{\$300{,}000n}{v^2} = 0.$$

So,

$$0.05625 v^{1/2} - \frac{300{,}000}{v^2} = 0$$

$$0.05625 v^{5/2} - 300{,}000 = 0$$

$$v^{5/2} = \frac{300{,}000}{0.05625} = 5{,}333{,}333$$

$$v^* = (5{,}333{,}333)^{0.4} = 490.68 \text{ mph}.$$

Chapter 2
Solution 2-6

How Fast Should the Airplane Fly?

(b) Finally, we check the second derivative to confirm a minimum cost solution:

$$\frac{d^2C_T}{dv^2} = \frac{0.028125}{v^{1/2}} + \frac{600{,}000}{v^3} \qquad \text{for } v > 0, \text{ and therefore, } \frac{d^2C_T}{dv^2} > 0.$$

The company concludes that $v = 490.68$ mph minimizes the total cost of this particular airplane's flight.

Chapter 2
Example

As energy costs continue to rise, power efficiency is increasingly important. Acme Chemical is evaluating two different electric motors to drive a mixing motor and needs to perform a present economy study.

The motor will produce 75 hp and will be operated eight hours per day, 365 days for one year (maintenance will be performed on second shift—assume no down time during operation), after which time the motor will have no value.

Select the most economical motor. Assume Acme's electric power costs \$0.16 per kWh. 1 hp = 0.746 kW.

	Motor A	Motor B
Purchase price	\$3,200	\$5,900
Annual maintenance cost	\$250	\$450
Efficiency	75%	85%

Chapter 2
Solution

Motor A

$$\left(\frac{75\text{hp}}{0.75}\right)\left(\frac{0.746\text{kW}}{\text{hp}}\right)\left(\frac{\$0.16}{\text{kWh}}\right)\left(\frac{8\text{hr}}{\text{day}}\right)\left(\frac{365\text{days}}{\text{year}}\right) = \$34,853$$

$$\text{Total Cost} = \$34,853 + \$3,200 = \$38,053$$

Motor B

$$\left(\frac{75\text{hp}}{0.85}\right)\left(\frac{0.746\text{kW}}{\text{hp}}\right)\left(\frac{\$0.16}{\text{kWh}}\right)\left(\frac{8\text{hr}}{\text{day}}\right)\left(\frac{365\text{days}}{\text{year}}\right) = \$30,753$$

$$\text{Total Cost} = \$30,753 + \$5,900 = \$36,653$$